

AI-Assisted Lending Experiment: A/B Testing and Statistical Evaluation

This project evaluated whether a newly developed AI-assisted loan review model improved loan officers' decision quality compared to an existing model. Using data from a controlled A/B experiment, I analysed the impact of the new model on Type I and Type II error rates, reviewer confidence, and decision-making behaviour. The analysis incorporated hypothesis testing, effect size measurement, and power analysis to assess both statistical significance and practical business impact.

Library Setup

```
knitr::opts_chunk$set(echo = TRUE)
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.4
## ✓ forcats    1.0.0      ✓ stringr   1.5.0
## ✓ ggplot2     3.5.1      ✓ tibble     3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr      1.3.0
## ✓ purrr       1.0.2
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
library(ggplot2)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##   group_rows
```

```
library(presenter)
```

Objective Overview:

The company has been reviewing loan applications and plans to conduct an A/B testing to see which computer model (Control: current computer model, Treatment: new computer model) performs best in helping to make better loan review decisions. The aim of this report is to discover which of the model minimizes the losses of the company from bad loans and maximize profit from good loans.

Read File: Loan Procedure

```
lr_df <- read_csv("ADAprject_2025_data.csv", show_col_types=FALSE)

#we have 22 columns of interest and 470 columns of entry
#total number of loans: 4700, we have imbalanced dataset in which bad loans is lower, which happens to a lot of real-life dataset.
#However, we are interested in the loan officer's decision quality overall, rather than the AI model itself

#setting the categorical variables
lr_df$Variant <- factor(lr_df$Variant)
lr_df$loanofficer_id <- factor(lr_df$loanofficer_id)

#number of rows in the dataset
nrow(lr_df)

## [1] 470
```

Initial Data Dictionary

Variable	Description
Variant	The experimental variant randomly assigned to each loan officer; Control and Treatment
loanofficer_id	Unique identifier for each loan officer
day	The day of the experiment (e.g. 1 means 1st day, 2 means 2nd day, etc.)
typel_init	Count of Each Loan officer's Type I errors (false positives – rejecting good loans) before seeing computer predictions

typel_fin	Count of Each Loan officer's Type I errors (false positives – rejecting good loans) after seeing computer predictions
typell_init	Count of Each Loan officer's Type II errors (false negatives – approving bad loans) before seeing computer predictions
typell_fin	Count of Each Loan officer's Type II errors (false negatives – approving bad loans) after seeing computer predictions
ai_typel	Count of computer model's Type I errors (false positives – rejecting good loans)
ai_typell	Count of computer model's Type II errors (false negatives – approving bad loans)
badloans_num	Number of bad loans (loans that defaulted)
goodloans_num	Number of good loans (loans that were paid back on time)
agree_init	Count of Each Loan officer's agreements with computer predictions before seeing computer predictions
agree_fin	Count of Each Loan officer's agreements with computer predictions after seeing computer predictions
conflict_init	Count of Each Loan officer's conflicts with computer predictions before seeing computer predictions
conflict_fin	Count of Each Loan officer's conflicts with computer predictions after seeing computer predictions
revised_per_ai	Count of Each Loan officer's decisions that were revised to follow computer predictions
revised_agst_ai	Count of Each Loan officer's decisions that were revised to go against computer predictions
confidence_init_total	Sum of confidence ratings given by each Loan Officer to their completed loan review decisions (how sure they were in their decisions) before seeing computer predictions
confidence_fin_total	Sum of confidence ratings given by each Loan Officer to their completed loan review decisions (how sure they were in their decisions) after seeing computer predictions
complt_init	Count of Initial loan review decisions completed by each Loan Officer before seeing computer predictions
complt_fin	Count of Final loan review decisions completed by each Loan Officer after seeing computer predictions
fully_complt	Count of each loan officer's fully completed loan reviews (in both stages – before and after seeing computer predictions)

Data Dictionary of Transformed Columns

Variable	Description
difference_typel	Difference in each loan officer's Type I error count (false positives - rejecting good loans) before versus after seeing computer predictions
difference_typell	Difference in each loan officer's Type II error count (false negatives - approving bad loans) before versus after seeing computer predictions
error_typel_percentage_change	Normalized change in Type I error difference, calculated as the difference in false positives before versus after seeing computer predictions, divided by actual total of non-default loans
error_typell_percentage_change	Normalized change in Type II error difference, calculated as the difference in false negatives before versus after seeing computer predictions, divided by actual total of default loans
officer_agree_difference	Difference in each loan officer's agreements with computer decisions before versus after seeing computer predictions
officer_conflict_difference	Difference in each loan officer's conflict with computer decisions before versus after seeing computer predictions
loan_confidence_difference	Difference in loan officer's confidence rating in their loan decisions before versus after seeing computer predictions

Data Check

```
#no duplicates found
sum(duplicated(lr_df))
```

```
## [1] 0
```

```
#checking for nulls: no nulls detected
sum(is.na(lr_df))
```

```
## [1] 0
```

```
#each loan officer is assigned to only one variant
(lr_df %>%
  group_by(loanofficer_id) %>%
  summarise(num_variants = n_distinct(Variant)) %>%
  filter(num_variants > 1))
```

```
## # A tibble: 0 × 2
## #   Variant    num_variants
## #   <fct>      <int>
```

```
#number of loan officers using one computer model: roughly equal distribution
loan_officer_proportion <-lr_df %>% group_by(Variant) %>% summarise(num_loan_officers = n_distinct(loanofficer_id),
                                                                    total_loan_officers = n_distinct(lr_df$loanofficer_id),
                                                                    officer_distribution_percentage = round(num_loan_officers/total_loan_officers*100),2)

loan_officer_proportion %>% select(Variant,num_loan_officers,officer_distribution_percentage)
```

```
## # A tibble: 2 × 3
##   Variant    num_loan_officers officer_distribution_percentage
##   <fct>          <int>                <dbl>
## 1 Control             19                     40
## 2 Treatment           28                     60
```

Total Loans Reviewed

- In total there are 4700 loans of which 3290 are non-default loans and 1410 are bad loans. This is an imbalanced dataset which may affect the detection of type II error
- 4451 out of 4700 loans are reviewed initially and only 3692 loans are reviewed after looking at the computer model. This difference between the initial and final review will be taken into account by filtering only loans that are both reviewed at both stages per loan officer.

```
#total loans reviewed initially: 4451
#indicates there are several loans not being investigated
#total loans reviewed after :3692, we should only looked where the loans reviewed initially and loans after is equal
#complt_fin <= badloans_num+good loans_num
#indicates there are several loans not being investigated

(total_loans_stat <- lr_df %>%
  summarize(total_loans = sum(badloans_num) + sum(goodloans_num),
            total_of_good_loans= sum(goodloans_num),
            total_of_bad_loans = sum(badloans_num),
            total_of_initial_review = sum(complt_init),
            total_of_final_review = sum(complt_fin)) %>% kbl() %>% kable_styling())
```

total_loans	total_of_good_loans	total_of_bad_loans	total_of_initial_review	total_of_final_review
4700	3290	1410	4451	3692

Completion of initial review decision is significantly lower in Control variant compared to treatment. This inevitably affects the sample size we will be conducting on the A/B experiment and thereby reducing the statistical power.

```
#all initial review decisions and confidence rating reviews were completed in both variants
(initial_review<-lr_df %>% group_by(Variant) %>% filter(confidence_init_total==0) %>% summarise(n=n()))
```

```
## # A tibble: 0 × 2
## #   Variant    n
## #   <fct>    <int>
```

```
(initial_review_count<-lr_df %>% group_by(Variant) %>% filter(complt_init==0) %>% summarise(n=n()))
```

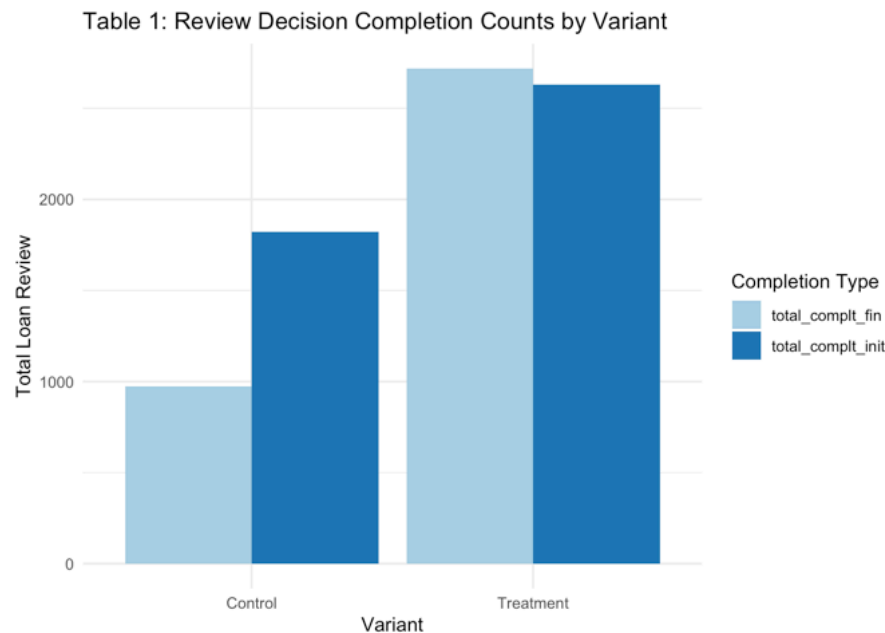
```
## # A tibble: 0 × 2
## #   Variant    n
## #   <fct>    <int>
```

```
#complt_init, and complt_fin

lr_review_decision <- lr_df %>% group_by(Variant) %>% summarise(total_complt_init = sum(complt_init),total_complt_fin = sum(complt_fin))

lr_review_long <- lr_review_decision %>% pivot_longer(cols = c(total_complt_init, total_complt_fin),
  names_to = "Completion_Type",
  values_to = "Total")

ggplot(lr_review_long, aes(x = Variant, y = Total, fill = Completion_Type)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Table 1: Review Decision Completion Counts by Variant",
    x = "Variant",
    y = "Total Loan Review",
    fill = "Completion Type") +
  scale_fill_brewer(palette = "Paired") + theme_minimal()
```



Control has 90 loans that are not reviewed of which we will proceed to remove from our EDA in addition to looking at loans that have both been reviewed before and after seeing the computer model.

```
# all final review control treatment has 90 loans not given confidence rating and not being reviewed as well
(final_review<-lr_df %>% group_by(Variant) %>% filter(confidence_fin_total==0) %>% summarise(n=n()))
```

```
## # A tibble: 1 × 2
##   Variant     n
##   <fct>   <int>
## 1 Control    90
```

```
(final_review_count<-lr_df %>% group_by(Variant) %>% filter(complt_fin==0) %>% summarise(n=n()))
```

```
## # A tibble: 1 × 2
##   Variant     n
##   <fct>   <int>
## 1 Control    90
```

Confusion Matrix of AI Model Evaluation Metric: Control versus Treamtent

After removing loans that have not been reviewed and finding only loans that been both reviewed before and after, we have 332 rows of data.

```
lr_df <- lr_df %>% filter(complt_init!=0, complt_fin!=0,complt_init==complt_fin)
nrow(lr_df)
```

```
## [1] 332
```

Accuracy,recall, and precision is substantially higher by using the treatment variant computer model.

```
#type I error (FP)
#type II error (FN)
lr_eval <- lr_df %>% group_by(Variant) %>% filter(complt_init!=0, complt_fin!=0,complt_init==complt_fin) %>%
  summarise(AI_FP=sum(ai_typeI)
            ,AI_FN=sum(ai_typeII)
            ,AI_TP=sum(badloans_num)-sum(ai_typeI)
            ,AI_TN=sum(goodloans_num)-sum(ai_typeII))

#treatment model is better across all evaluation metrics: accuracy, precision, and recall
#precision is still quite poor after using the treatment variant
#recall however is quite good
lr_eval %>% group_by(Variant) %>% summarise(accuracy = round((AI_TP+AI_TN) / (AI_FP+AI_FN+AI_TP+AI_TN),3),precision = round((AI_TP) / (AI_FP+AI_TP),3),recall = round((AI_TP) / (AI_FN+AI_TP),3)) %>% kbl() %>% kable_styling()
```

Variant	accuracy	precision	recall
Control	0.609	0.243	0.324
Treatment	0.800	0.544	0.702

Before versus After Loan Officer's Error Rate

```
#data transformation: before vs after difference loan officer decision making process
#after - before calculation
lr_df$difference_typeI <- lr_df$typeI_fin - lr_df$typeI_init
lr_df$difference_typeII <- lr_df$typeII_fin - lr_df$typeII_init
lr_df$officer_agree_difference <- lr_df$agree_fin - lr_df$agree_init
lr_df$officer_conflict_difference <- lr_df$conflict_fin - lr_df$conflict_init
lr_df$loan_confidence_difference<- lr_df$confidence_fin_total - lr_df$confidence_init_total
```

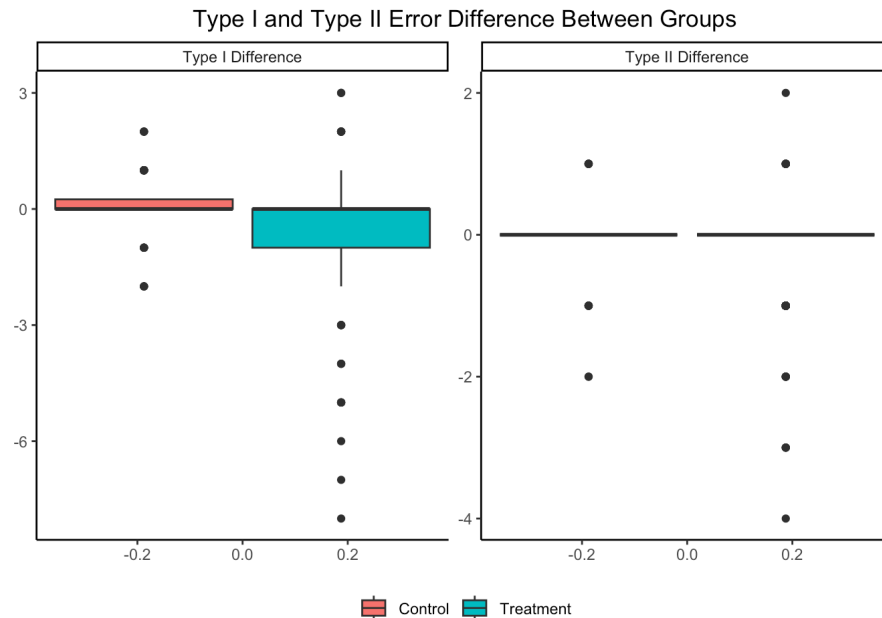
Boxplot below indicates that the difference in type I and type II errors by variants have little to no difference between control and treatment variant as they both have the same median = 0. However, we can observe that the treatment model displays more significant outliers.

```
#Data Summary of Type I and Type II Error
(summed_type_error <- lr_df %>%
  filter(complt_init!=0, complt_fin!=0,complt_init==complt_fin) %>%
  group_by(Variant) %>%
  summarise(
    total_typeI_errors_perc = (sum(typeI_fin)-sum(typeI_init))/sum(goodloans_num)*100
    ,total_typeII_errors_perc = (sum(typeII_fin)-sum(typeII_init))/sum(badloans_num)*100))%>% kbl() %>% kable_styling()
  ling())
```

Variant	total_typeI_errors_perc	total_typeII_errors_perc
Control	1.973684	-1.102941
Treatment	-5.603697	-8.744711

```
#variance in treatment for type I and type 2 error significantly reduced, although the median remains that there is no difference between the before and after type error rate

lr_df_new <- lr_df %>%
  mutate(diff_typeI = typeI_fin-typeI_init,
         diff_typeII = typeII_fin - typeII_init) %>%
  select(Variant,day,loanofficer_id,diff_typeI,diff_typeII)
lr_df_long <- lr_df_new %>%
  pivot_longer(cols = c('diff_typeI','diff_typeII'),names_to = 'type',values_to = 'values')
lr_df_long$type <- factor(lr_df_long$type, levels = c("diff_typeI", "diff_typeII"),labels=c('Type I Difference','Type II Difference'))
lr_df_long %>%
  ggplot(aes(y=values,fill=Variant))+geom_boxplot()+
  facet_wrap(~type,scales='free_y')+theme_classic()+labs(y=NULL,title = 'Type I and Type II Error Difference Between Groups',fill=NULL)+
  theme(legend.position = 'bottom',
        plot.title = element_text(hjust=0.5))
```



In terms of proportion of the total review decisions we see that for the treatment variant there is a small reduction in both Type I and Type II error. In comparison, the control variate only reduces type II error but increases type I error.

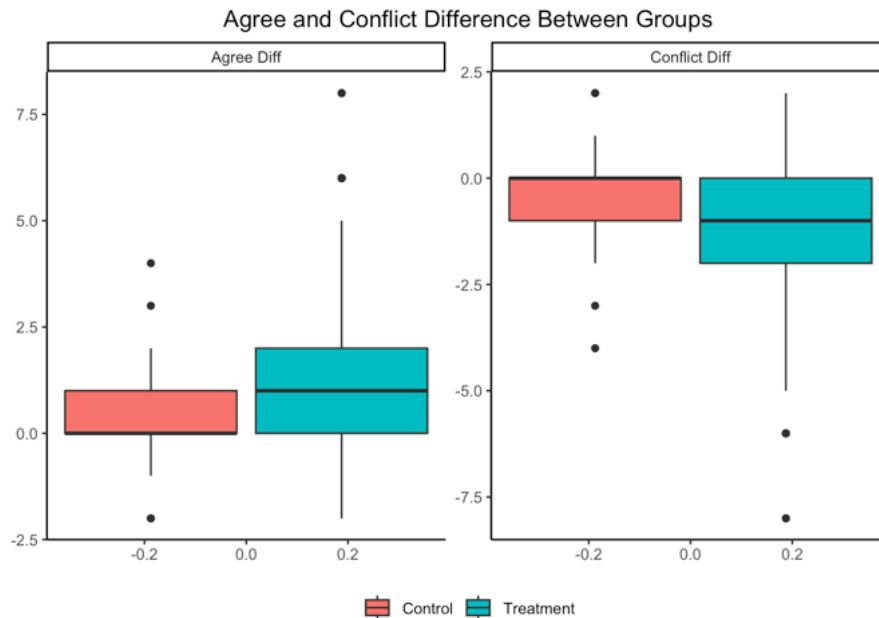
Agreement and Conflict Metrics

Loan officers make the final decision based on computer prediction model. As observed in the boxplot below, we see that agreement is higher by using the treatment variant and less conflict between loan officer decision and computer prediction model

```
#agreements higher and conflict is lower after using treatment computer model
lr_df %>%
  filter(complt_init!=0, complt_fin!=0,complt_init==complt_fin) %>%
  group_by(Variant) %>% summarise(average_agree_model = round(mean(officer_agree_difference/complt_fin),4), average_conflict_model=round(mean(officer_conflict_difference/complt_fin),4)) %>% kbl() %>% kable_styling()
```

Variant	average_agree_model	average_conflict_model
Control	0.0393	-0.0393
Treatment	0.1102	-0.1102

```
#boxplot however showing high variability in the treatment computer model in terms of agreement and conflict metric
lr_df_new <- lr_df %>%
  mutate(diff_agree = agree_fin-agree_init,
         diff_conflict = conflict_fin - conflict_init) %>%
  select(Variant,day,loanofficer_id,diff_agree,diff_conflict)
lr_df_long <- lr_df_new %>%
  pivot_longer(cols = c('diff_agree','diff_conflict'),names_to = 'type',values_to = 'values')
lr_df_long$type <- factor(lr_df_long$type, levels = c("diff_agree", "diff_conflict"),labels=c('Agree Diff','Conflict Diff'))
lr_df_long %>%
  ggplot(aes(y=values,fill=Variant))+geom_boxplot()+
  facet_wrap(~type,scales='free_y')+theme_classic()+labs(y=NULL,title = 'Agree and Conflict Difference Between Groups',fill=NULL)+
  theme(legend.position = 'bottom',
        plot.title = element_text(hjust=0.5))
```



Decision Revision Metrics

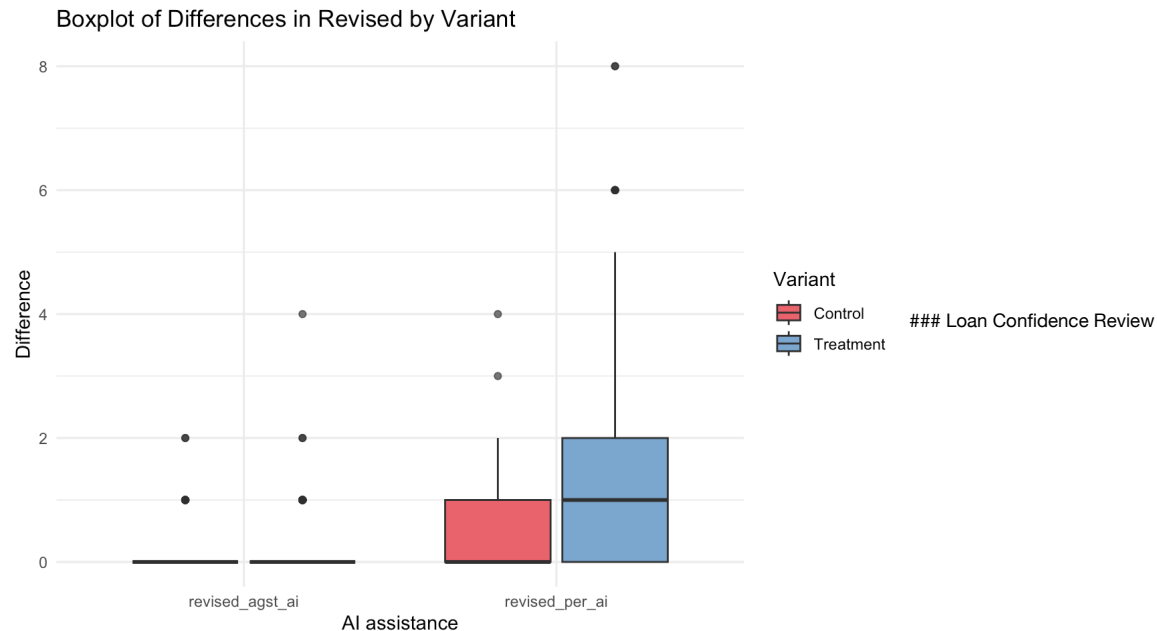
Additionally, by using the treatment variant model, we see that loan officers generally follow the computer model more compared to using the control variant and additionally we see a drop in revising decision against the computer model's decision in the treatment variant.

```
lr_df %>%
  filter(complt_init!=0, complt_fin!=0,complt_init==complt_fin) %>%
  group_by(Variant) %>%
  summarise(follow_ai_percent=sum(revised_per_ai)/sum(complt_fin)*100,against_ai_percent=sum(revised_agst_ai)/sum(complt_fin)*100) %>% kbl() %>% kable_styling()
```

Variant	follow_ai_percent	against_ai_percent
Control	5.485714	1.7142857
Treatment	11.926229	0.9016393

```
lr_df_long_3 <- lr_df %>% filter(complt_init!=0, complt_fin!=0,complt_init==complt_fin) %>%
pivot_longer(cols = starts_with("revised_")
              ,names_to="AI_assistance"
              ,values_to = "AI_count_revised")

#Based on the plot below we see high variability in revised per ai when officers use the treatment computer model
ggplot(lr_df_long_3, aes(x = AI_assistance, y = AI_count_revised, fill = Variant)) +
  geom_boxplot(alpha = 0.7) +
  labs(
    x = "AI assistance",
    y = "Difference",
    title = "Boxplot of Differences in Revised by Variant",
    fill = "Variant"
  ) +
  theme_minimal() + scale_fill_brewer(palette = "Set1")
```



Ratings and Decision Complete Counts

On average, confidence review is significantly higher in treatment variant (76%) compared to the control variant (61%).

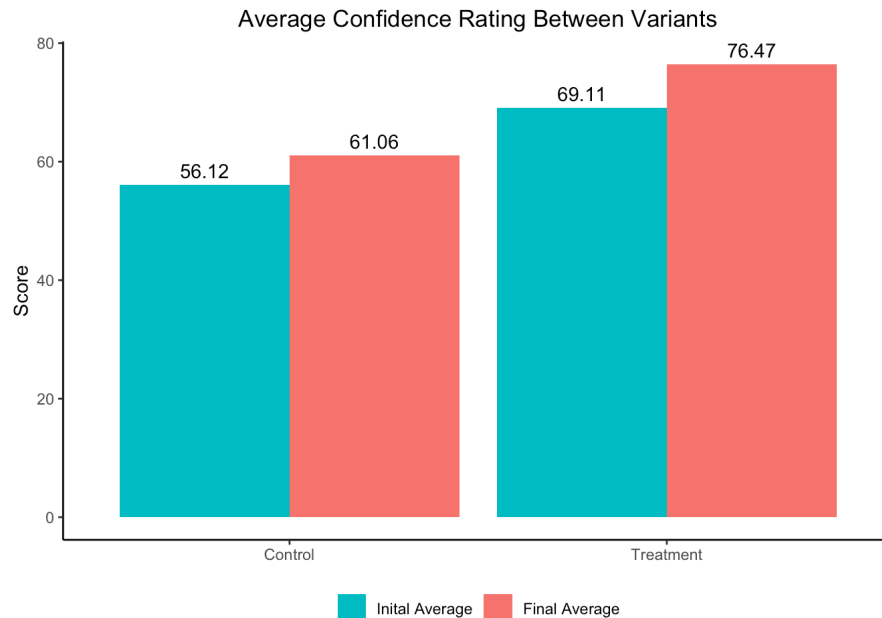
```
#before versus after
#shows officers are more confident in the treatment model and more sure of their decision after seeing the treatment computer predictions

lr_df <- lr_df %>% filter(complt_init==complt_fin,complt_init!=0, complt_fin!=0)

#based on the summary table we see that confidence rate significantly increases with the treatment model on average
(average_confidence<-lr_df %>% group_by(Variant) %>% summarise(average_initial_confidence=round(sum(confidence_init_total)/sum(complt_init),2)
                                                                    ,average_final_confidence = round(sum(confidence_fin_total)/sum(complt_fin),2)) %>%
  pivot_longer(cols=starts_with("average_")
               ,names_to = ("Confidence_Average")
               ,values_to = ("Average_Confidence")) %>% kbl() %>% kable_styling()
```

Variant	Confidence_Average	Average_Confidence
Control	average_initial_confidence	56.12
Control	average_final_confidence	61.06
Treatment	average_initial_confidence	69.11
Treatment	average_final_confidence	76.47

```
lr_df_new <- lr_df |>
  select(Variant,day,loanofficer_id,confidence_init_total,confidence_fin_total,complt_init,complt_fin)
lr_df_new <- lr_df_new |>
  group_by(Variant) |>
  summarise(average_init = sum(confidence_init_total)/sum(complt_init),
            average_fin = sum(confidence_fin_total)/sum(complt_fin))
lr_df_new <- lr_df_new |>
  pivot_longer(cols=c('average_init','average_fin'),names_to = 'type',values_to = 'values')
lr_df_new$type <- factor(lr_df_new$type, levels = c("average_init", "average_fin"),labels=c('Initial Average','Final Average'))
lr_df_new |>
  ggplot(aes(x=Variant,y=values,fill=type,label=round(values,2)))+geom_col(position = 'dodge')+geom_text(position =position_dodge(width=0.9), vjust=-0.5)+
  theme_classic()+labs(fill=NULL,y='Score',x=NULL,title='Average Confidence Rating Between Variants')+
  scale_fill_manual(values = c("Initial Average" = "#00BFC4", "Final Average" = "#F8766D"))+
  theme(plot.title = element_text(hjust=.5),
        legend.position = 'bottom')
```

Overall Evaluation Criteria (OEC)

Task: analyse the experiment whether exposing loan officers to the new computer model helped reduced the type I and type II error rates (helped improved their decision making process).

Overall Evaluated Criteria (Weighted) = $w_1 \times \text{Type I Error Percentage Change Per Loan Officer} + w_2 \times \text{Type II Error Percentage Change Per Loan Officer}$

For the purpose of this analysis, we will let $w_1 = 0.4$ and $w_2 = 0.6$ given the impact of Type II Errors being financially riskier compared to Type I Error for the company.

```
officer_level_data <- lr_df %>%
  filter(complt_init!=0, complt_fin!=0,complt_init==complt_fin) %>%
  group_by(Variant,loanofficer_id) %>%
  mutate(error_type_I_percentage_change=difference_typeI/sum(goodloans_num)
          ,error_type_II_percentage_change=difference_typeII/sum(badloans_num)
          ,.groups = "drop")

head(officer_level_data)
```

```
## # A tibble: 6 × 30
## # Groups:   Variant, loanofficer_id [1]
##   Variant loanofficer_id day typeI_init typeI_fin typeII_init typeII_fin
##   <fct>    <fct>         <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 Treatment qamcqdoe         2         2         2         3         3
## 2 Treatment qamcqdoe         3         3         3         0         0
## 3 Treatment qamcqdoe         4         1         2         1         1
## 4 Treatment qamcqdoe         5         0         2         0         0
## 5 Treatment qamcqdoe         6         0         1         4         0
## 6 Treatment qamcqdoe         7         0         1         1         0
## # i 23 more variables: agree_init <dbl>, agree_fin <dbl>, conflict_init <dbl>,
## #   conflict_fin <dbl>, revised_per_ai <dbl>, revised_agst_ai <dbl>,
## #   fully_complt <dbl>, confidence_init_total <dbl>,
## #   confidence_fin_total <dbl>, complt_init <dbl>, complt_fin <dbl>,
## #   ai_typeI <dbl>, ai_typeII <dbl>, badloans_num <dbl>, goodloans_num <dbl>,
## #   difference_typeI <dbl>, difference_typeII <dbl>,
## #   officer_agree_difference <dbl>, officer_conflict_difference <dbl>, ...
```

```
#there are 2 rows that do not have 10 loan reviews completed in initial and final
nrow(officer_level_data %>% filter(complt_fin!=10 & complt_init!=10))
```

```
## [1] 2
```

```
#there are 330 rows (majority of rows have both 10 loans reviewed in the initial and final stage)
nrow(officer_level_data %>% filter(complt_fin==10 & complt_init==10))
```

```
## [1] 330
```

```
officer_level_statistics<-officer_level_data %>%
  group_by(Variant) %>%
  summarise(
    mean_typeII_percentage_change = mean(error_type_II_percentage_change)
    ,var_typeII_percentage_change = var(error_type_II_percentage_change)
    ,mean_typeI_percentage_change = mean(error_type_I_percentage_change)
    ,var_typeI_percentage_change = var(error_type_I_percentage_change)
    ,n = n()
  )

#loan range: [20,000 - 35000]
loan_range <- c(20000,35000)

officer_level_statistics
```

```
## # A tibble: 2 × 6
##   Variant    mean_typeII_percenta...1 var_typeII_percentag...2 mean_typeI_percentag...3
##   <fct>          <dbl>          <dbl>          <dbl>
## 1 Control          -0.00124          0.000345          0.00210
## 2 Treatment        -0.0101          0.000767         -0.00644
## # i abbreviated names: 1mean_typeII_percentage_change,
## # 2var_typeII_percentage_change, 3mean_typeI_percentage_change
## # i 2 more variables: var_typeI_percentage_change <dbl>, n <int>
```

OEC Decision Quality Analysis: Loss Reduction

#greater negative values indicates greater reduction loss given that we are measuring the difference before versus after, per 100 loans to simplify the financial impact

```
total_saving_cost <- lr_df %>%
  group_by(Variant) %>%
  summarise(
    reduction_loss_type_I_min = (sum(difference_typeI) / sum(goodloans_num)) * 100 * loan_range[1],
    reduction_loss_type_I_max = (sum(difference_typeI) / sum(goodloans_num)) * 100 * loan_range[2],
    reduction_loss_type_II_min = (sum(difference_typeII) / sum(badloans_num)) * 100 * loan_range[1], # Fixed mis
    placed parentheses
    reduction_loss_type_II_max = (sum(difference_typeII) / sum(badloans_num)) * 100 * loan_range[2],
    net_reduction_loss = (reduction_loss_type_I_max + reduction_loss_type_II_max) - (reduction_loss_type_I_min -
    reduction_loss_type_I_min) # Corrected error
  )

# Display the normalized value by using the expected value approach
expected_value<-total_saving_cost %>% group_by(Variant) %>%
  summarise(
    type_I_expected_financial_exposure= (reduction_loss_type_I_min+reduction_loss_type_I_max)/2,
    type_II_expected_financial_exposure = (
      reduction_loss_type_II_min+reduction_loss_type_II_max)/2,
    total_expected_exposure = type_I_expected_financial_exposure + type_II_expected_financial_exposure
  )

expected_value%>% kbl() %>% kable_styling()
```

Variant	type_I_expected_financial_exposure	type_II_expected_financial_exposure	total_expected_exposure
Control	54276.32	-30330.88	23945.43
Treatment	-154101.68	-240479.55	-394581.22

```
OEC_metric <- expected_value %>%
  summarise(type_I_OEC_percentage_change = (type_I_expected_financial_exposure[Variant=="Treatment"] - type_I_exp
  ected_financial_exposure[Variant=="Control"] )/type_I_expected_financial_exposure[Variant=="Control"],
    type_II_OEC_percentage_change = (type_II_expected_financial_exposure[Variant=="Treatment"] - type_II_
  expected_financial_exposure[Variant=="Control"] )/type_II_expected_financial_exposure[Variant=="Control"],
    OEC_result = 0.4*type_I_OEC_percentage_change + 0.6*type_II_OEC_percentage_change) %>% kbl() %>% kabl
  e_styling()
```

```
lr_df %>%
  filter(complt_init!=0, complt_fin!=0,complt_init==complt_fin) %>%
  group_by(Variant) %>%
  summarise(mean_type_I_before = mean(typeI_init)
            ,sd_type_I_before = sd(typeI_init)
            ,mean_type_I_after = mean(typeI_fin)
            ,sd_type_I_after = sd(typeI_fin)
            ,mean_type_II_before = mean(typeII_init)
            ,sd_type_II_before = sd(typeII_init)
            ,mean_type_II_after = mean(typeII_fin)
            ,sd_type_II_after = sd(typeII_fin)
            ,n = n()) %>% kbl() %>% kable_styling()
```

Variant	mean_type_I_before	sd_type_I_before	mean_type_I_after	sd_type_I_after	mean_type_II_before	sd_type_II_before	mean_type_II_after
Control	3.488636	1.695120	3.625000	1.656145	1.204546	1.204962	1.17041
Treatment	2.418033	1.840738	2.020492	1.397967	1.020492	1.059734	0.76631

#Normalised t-test based on completed review loans

```
# part ii
#welch t-test: given that the variance is unequal
officer_level_data <- officer_level_data %>% mutate(error_type_I_percentage_change = difference_typeI/sum(goodloans_num)
                                                    ,error_type_II_percentage_change = difference_typeII/sum(badloans_num))

improvement_rate_per_variant_per_loan <-officer_level_data %>% group_by(Variant) %>% summarise(mean(error_type_I_percentage_change),mean(error_type_II_percentage_change))

improvement_rate_per_variant_per_loan %>% kbl() %>% kable_styling()
```

Variant	mean(error_type_I_percentage_change)	mean(error_type_II_percentage_change)
Control	0.0020952	-0.0012392
Treatment	-0.0064390	-0.0101072

```
#unequal variance fulfilled to perform Welch two sample t-test
#1. variance unequal between treatment and control
#2. Each loan officer is subjected to one variant of the A/B test, fulfilling independent criteria (this is fulfilled already based on the code above)

#check for variance assumption is fulfilled
print(var(officer_level_data$error_type_I_percentage_change))
```

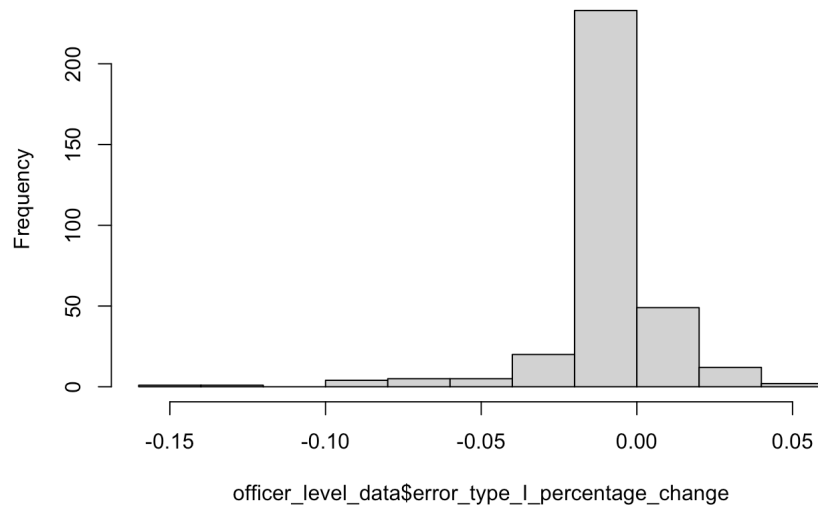
```
## [1] 0.0004926192
```

```
print(var(officer_level_data$error_type_II_percentage_change))
```

```
## [1] 0.0006691941
```

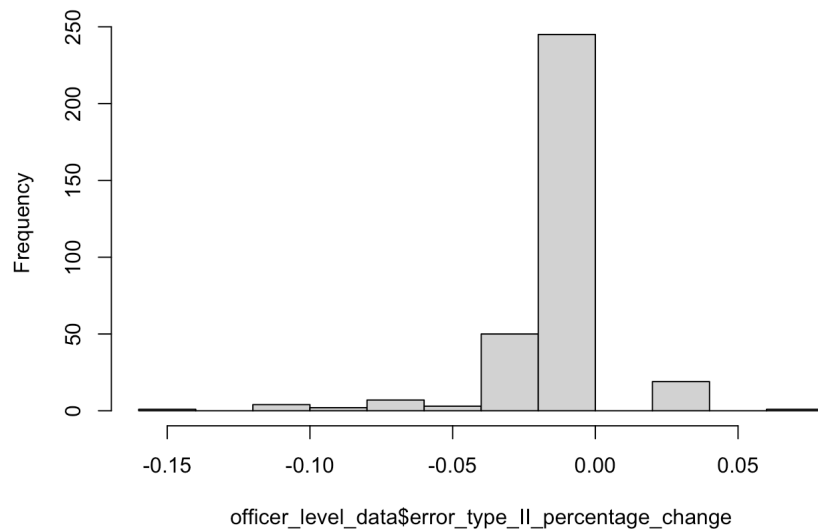
```
#check for normality assumption: distribution appears to be normal, slightly skewed. Normality Assumption is fulfilled.
hist(officer_level_data$error_type_I_percentage_change)
```

Histogram of officer_level_data\$error_type_I_percentage_change



```
hist(officer_level_data$error_type_II_percentage_change)
```

Histogram of officer_level_data\$error_type_II_percentage_change



```
# Type I error: significant, treatment significantly reduced type I error difference per loan officer compared to control
t.test(
  error_type_I_percentage_change~ Variant
, data = officer_level_data,
var.equal = FALSE)
```

```
##
## Welch Two Sample t-test
##
## data: error_type_I_percentage_change by Variant
## t = 4.1495, df = 291.7, p-value = 4.376e-05
## alternative hypothesis: true difference in means between group Control and group Treatment is not equal to 0
## 95 percent confidence interval:
## 0.004486379 0.012582009
## sample estimates:
## mean in group Control mean in group Treatment
## 0.002095208 -0.006438986
```

```
# Type II error:significant, treatment model significantly reduced type II error difference per loan officer compared to control
t.test(
  error_type_II_percentage_change~ Variant
  ,data = officer_level_data,
  var.equal = FALSE)
```

```
##
## Welch Two Sample t-test
##
## data: error_type_II_percentage_change by Variant
## t = 3.3369, df = 229.67, p-value = 0.0009881
## alternative hypothesis: true difference in means between group Control and group Treatment is not equal to 0
## 95 percent confidence interval:
## 0.003631666 0.014104410
## sample estimates:
## mean in group Control mean in group Treatment
## -0.001239177 -0.010107216
```

```
# Compute mean OEC for each Variant
mean_OEC_each_Variant_type_I_error <- officer_level_data %>%
  group_by(Variant) %>%
  summarise(mean_type_I_error_percentage_change = mean(error_type_I_percentage_change))

mean_OEC_each_Variant_type_II_error <- officer_level_data %>%
  group_by(Variant) %>%
  summarise(mean_type_II_error_percentage_change = mean(error_type_II_percentage_change))

# View mean OEC
#improvement by using treatment for type I and type II error
print(mean_OEC_each_Variant_type_I_error)
```

```
## # A tibble: 2 × 2
##   Variant mean_type_I_error_percentage_change
##   <fct>                                <dbl>
## 1 Control                                0.00210
## 2 Treatment                            -0.00644
```

```
print(mean_OEC_each_Variant_type_II_error)
```

```
## # A tibble: 2 × 2
##   Variant mean_type_II_error_percentage_change
##   <fct>                                <dbl>
## 1 Control                                -0.00124
## 2 Treatment                            -0.0101
```

```
# Compute pairwise % differences in OEC between pairs of variants
pairwise_diff_type_I <- mean_OEC_each_Variant_type_I_error %>%
  summarise(
    Diff_T_Control = mean_type_I_error_percentage_change[Variant== "Treatment"]- mean_type_I_error_percentage_change[Variant== "Control"]
    ,Perc_T_Control = (Diff_T_Control / mean_type_I_error_percentage_change[Variant== "Control"]) * 100
  )

pairwise_diff_type_2 <- mean_OEC_each_Variant_type_II_error %>%
  summarise(
    Diff_T_Control = mean_type_II_error_percentage_change[Variant== "Treatment"]- mean_type_II_error_percentage_change[Variant== "Control"]
    ,Perc_T_Control = (Diff_T_Control / mean_type_II_error_percentage_change[Variant== "Control"]) * 100
  )

print(pairwise_diff_type_I)
```

```
## # A tibble: 1 × 2
##   Diff_T_Control Perc_T_Control
##   <dbl>          <dbl>
## 1      -0.00853      -407.
```

```
print(pairwise_diff_type_2)
```

```
## # A tibble: 1 × 2
##   Diff_T_Control Perc_T_Control
##   <dbl>          <dbl>
## 1      -0.00887      716.
```

```
officer_level_data %>% group_by(Variant) %>% summarise(sample_size=n())
```

```
## # A tibble: 2 × 2
##   Variant    sample_size
##   <fct>         <int>
## 1 Control             88
## 2 Treatment          244
```

Cohen's d to measure effect size

```
#Cohen's d test on type I error percentage change
Control = officer_level_data$error_type_I_percentage_change[officer_level_data$Variant== "Control"]
Trt_I = officer_level_data$error_type_I_percentage_change[officer_level_data$Variant== "Treatment"]
library(effectsize)

cohens_d(Trt_I, Control)
```

```
## Cohen's d |          95% CI
## -----
## -0.39      | [-0.63, -0.14]
##
## - Estimated using pooled SD.
```

```
#cohen's d test on type II error percentage change
Control = officer_level_data$error_type_II_percentage_change[officer_level_data$Variant== "Control"]
Trt_II = officer_level_data$error_type_II_percentage_change[officer_level_data$Variant== "Treatment"]
library(effectsize)

cohens_d(Trt_II, Control)
```

```
## Cohen's d |          95% CI
## -----
## -0.35      | [-0.59, -0.10]
##
## - Estimated using pooled SD.
```

Power Test: Sample Size Requirement for Both Variants

```
# we need 104 samples for both control and treatment. While treatment satisfies the threshold set by the cohen's
d, control sample size is under the value needed
library(pwr)
pwr.t.test(
  power = 0.8,
  d = 0.39,
  sig.level = 0.05,
  type = "two.sample")
```

```
##
##   Two-sample t test power calculation
##
##           n = 104.1757
##           d = 0.39
##       sig.level = 0.05
##           power = 0.8
##   alternative = two.sided
##
## NOTE: n is number in *each* group
```

```
# once again the treatment sample size satisfies the necessary power needed however control sample size is still below, causing an imbalance which affects the Welch t-test reliability in terms of the significance
pwr.t.test(
  power = 0.8,
  d = 0.35,
  sig.level = 0.05,
  type = "two.sample")
```

```
##
##      Two-sample t test power calculation
##
##              n = 129.1121
##              d = 0.35
##      sig.level = 0.05
##              power = 0.8
##      alternative = two.sided
##
## NOTE: n is number in *each* group
```